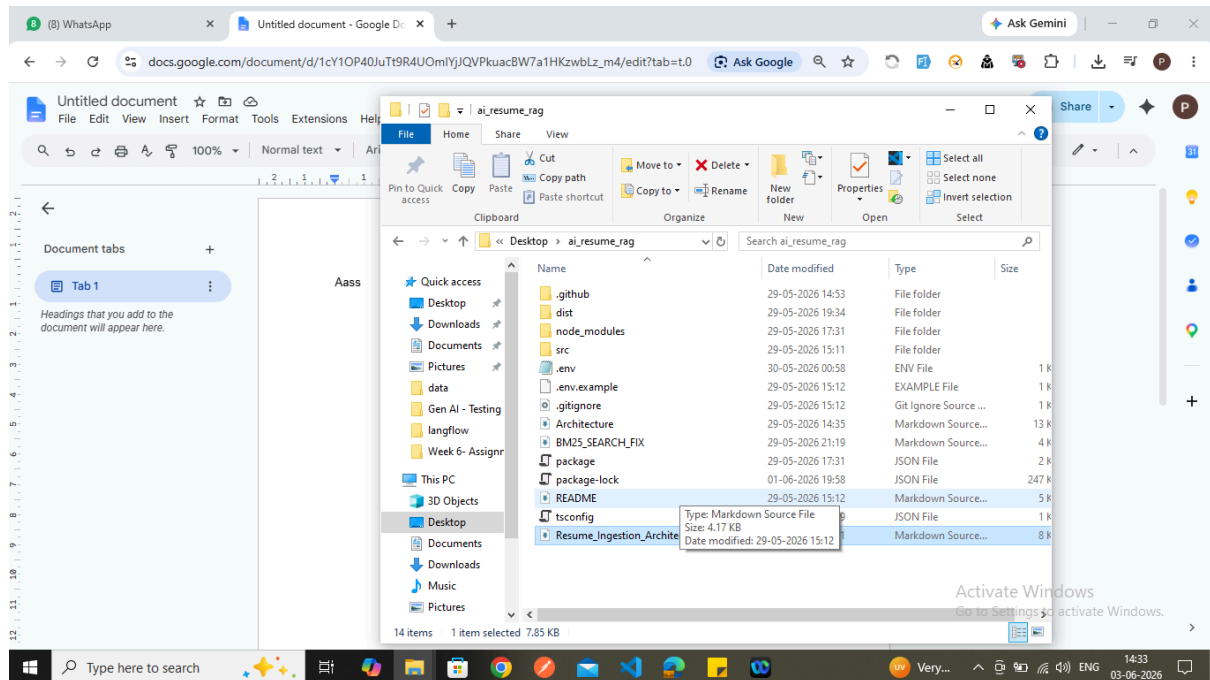
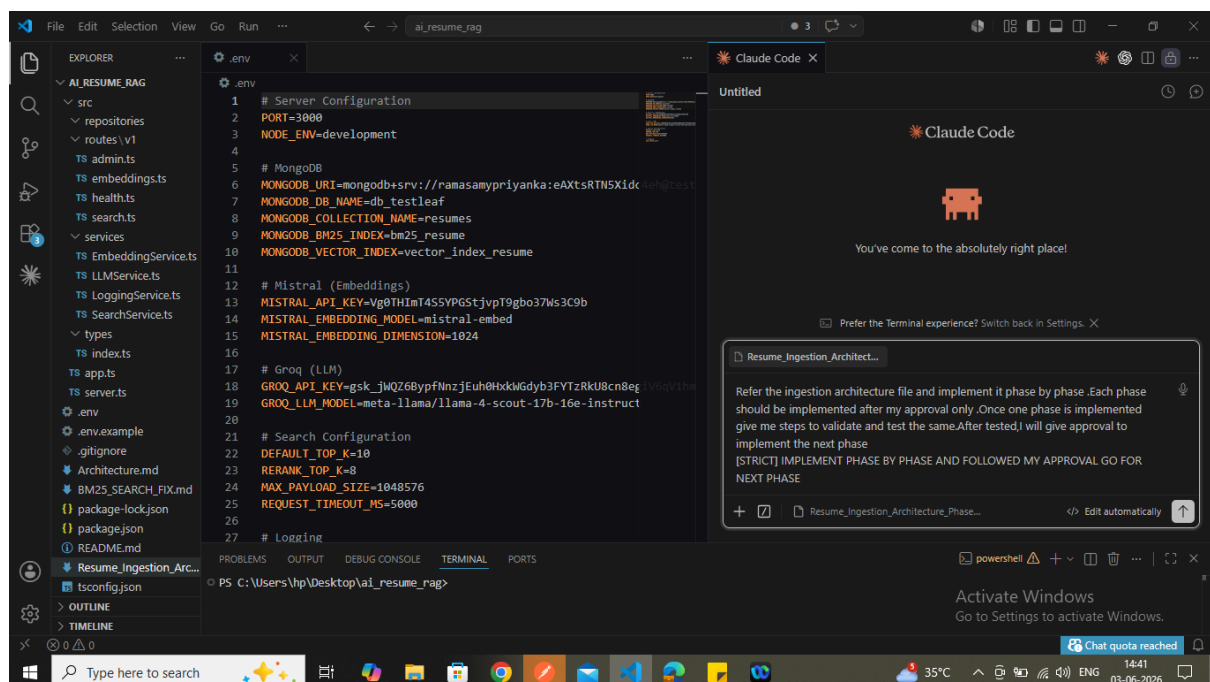


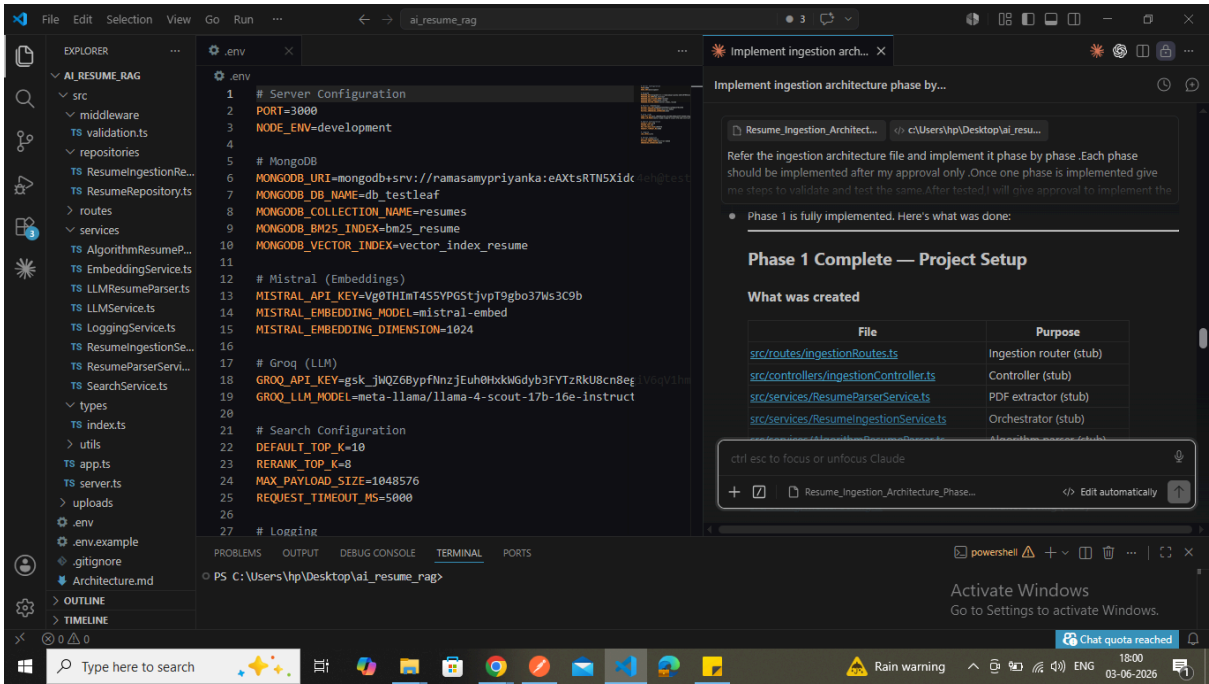
## Add Resume\_Ingestion Pipeline-Implementation



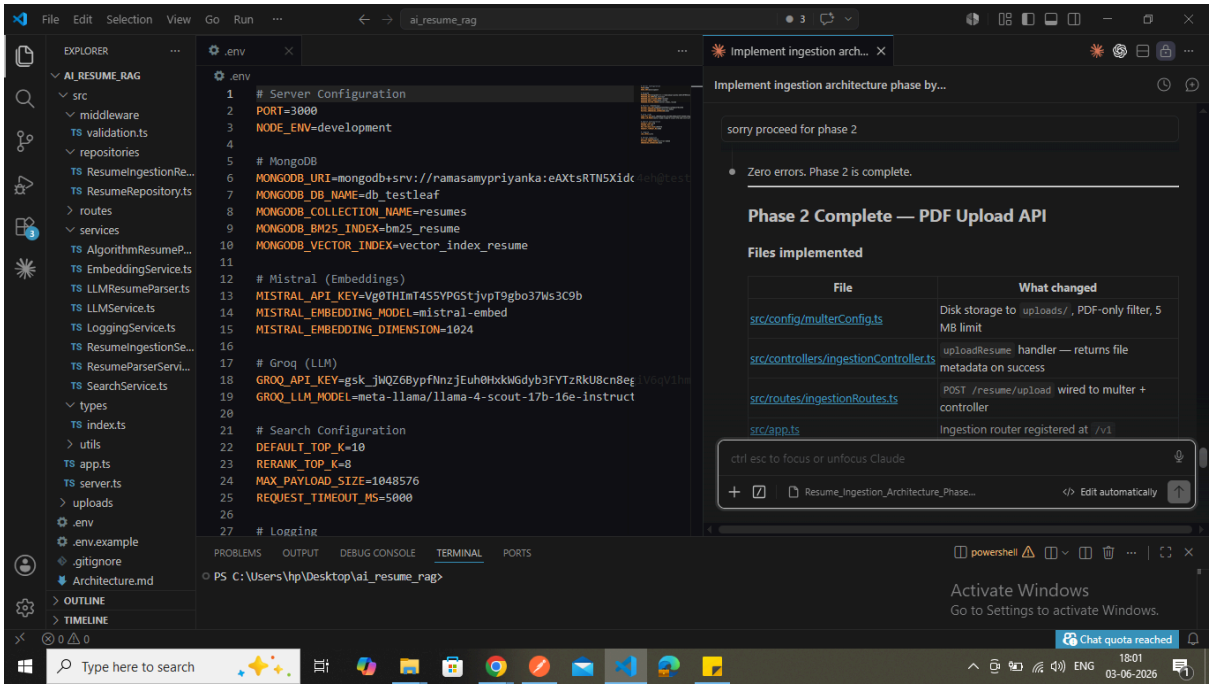
Refer the ingestion architecture file and implement it phase by phase .Each phase should be implemented after my approval only .Once one phase is implemented give me steps to validate and test the same.After tested,I will give approval to implement the next phase [STRICT] IMPLEMENT PHASE BY PHASE AND FOLLOWED MY APPROVAL GO FOR NEXT PHASE



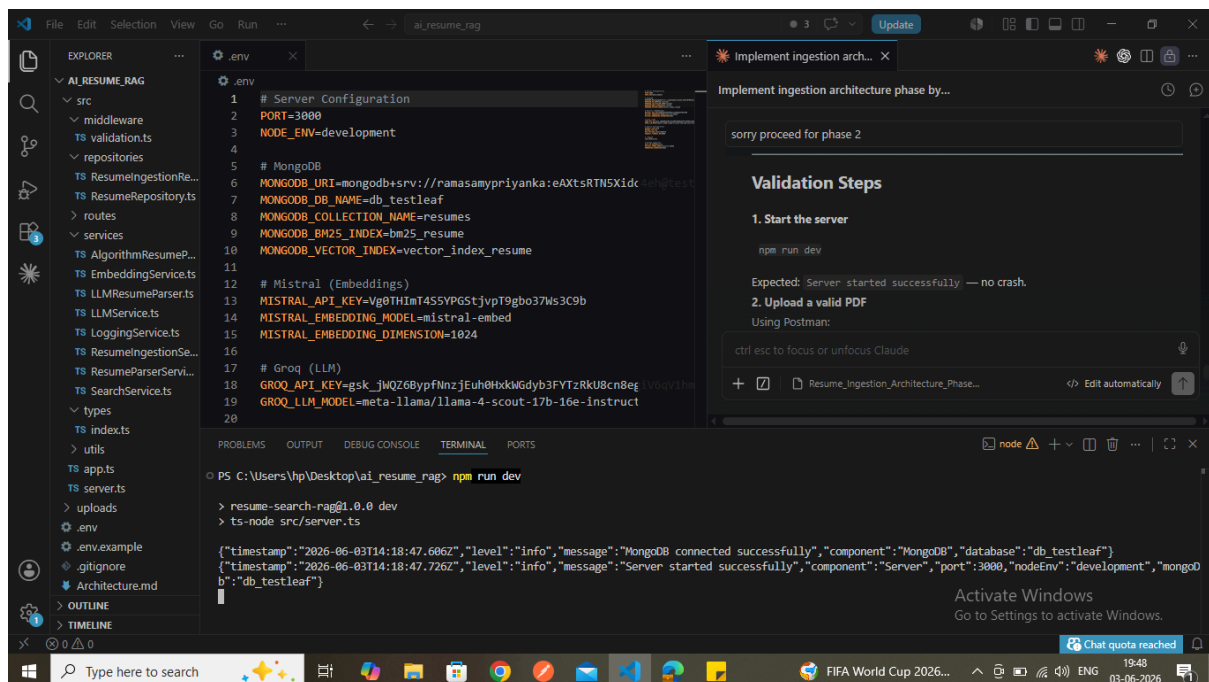
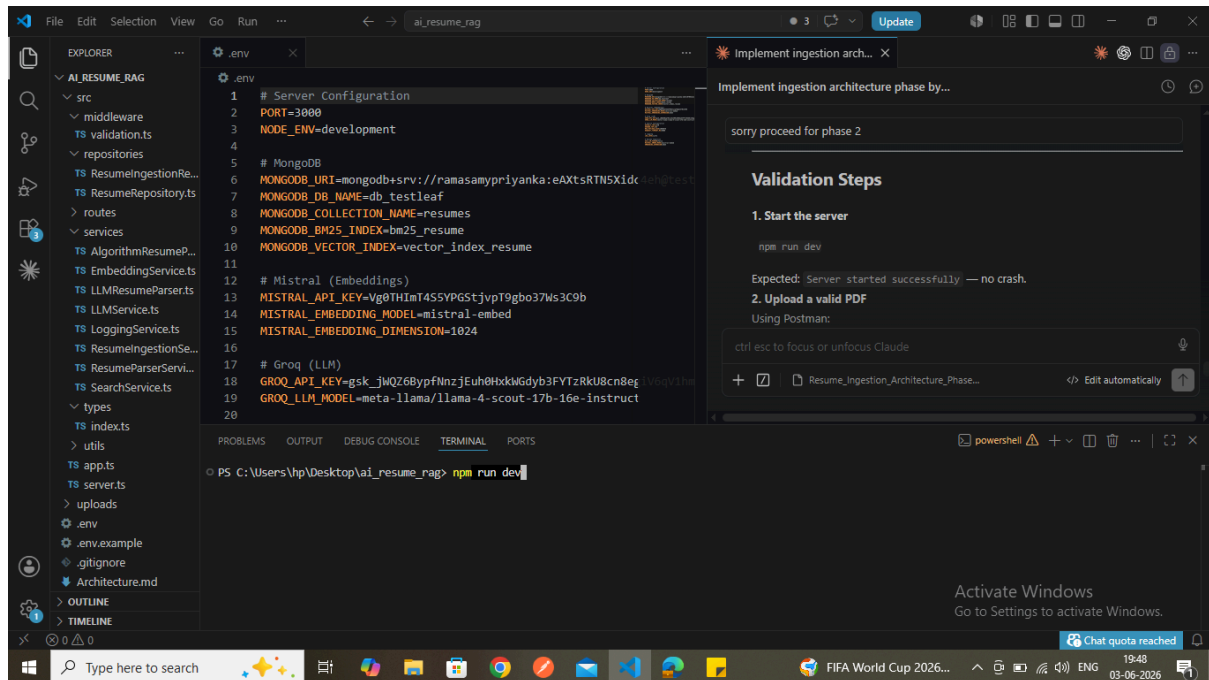
Phase 1 is completed



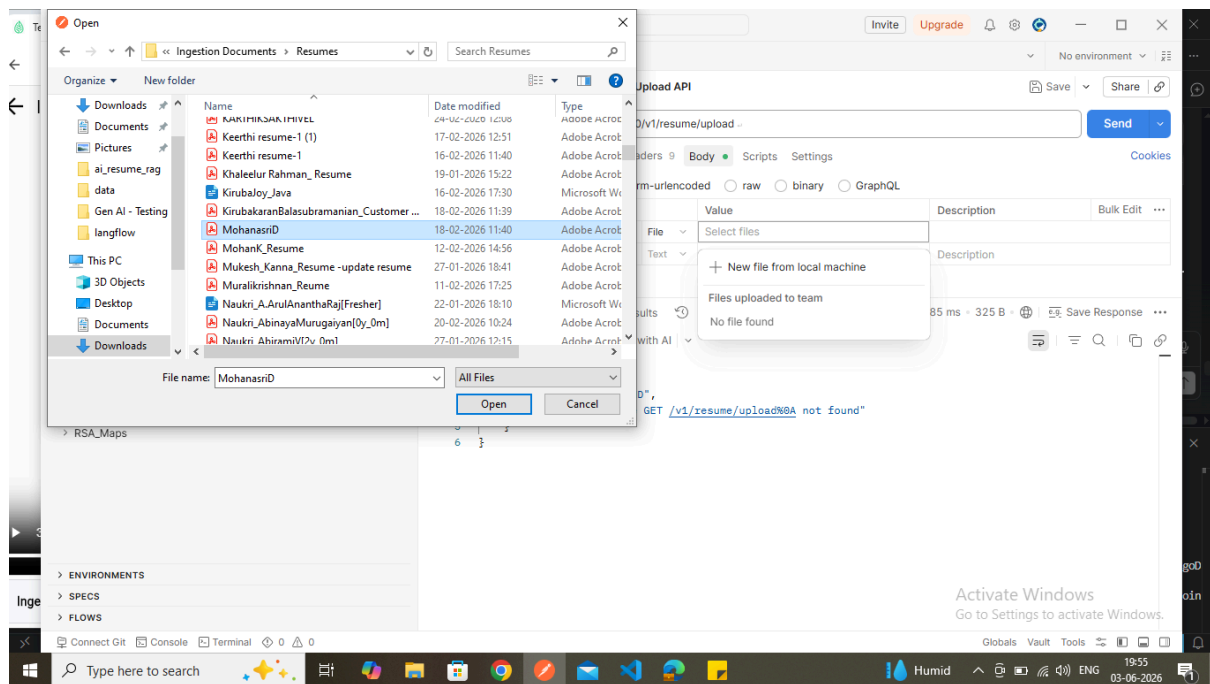
Phase 2 - PDF Upload API is completed



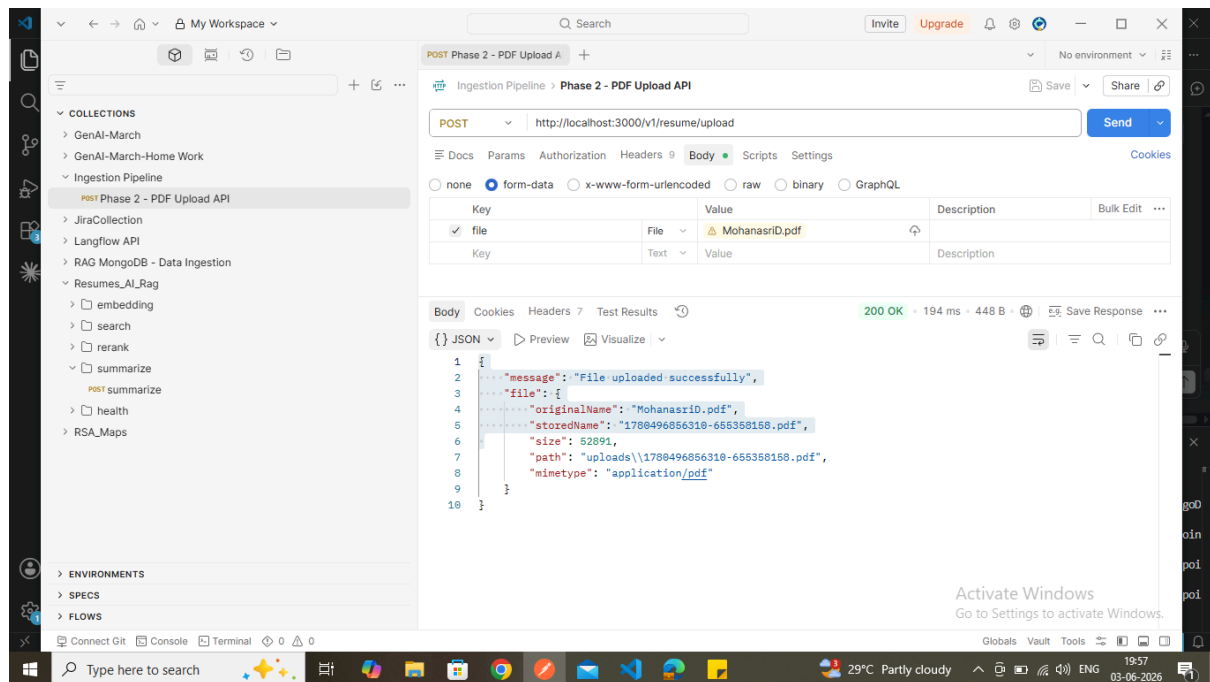
## Validating PDF Upload via API



Created New Collection and added the request and uploaded pdf files



Sent an request in postman and file is uploaded successfully



Uploaded .docx resume and it thrown an error

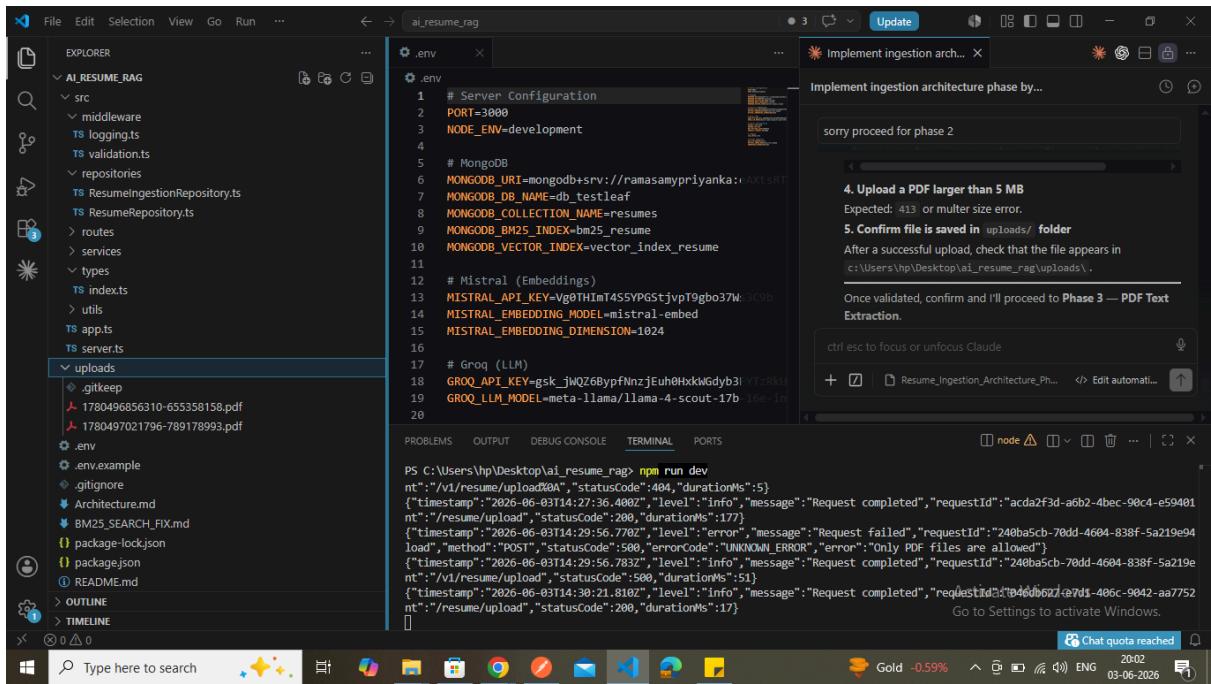
The top screenshot shows a file explorer window with the path 'Ingestion Documents > Resumes'. A list of files is displayed, including 'Akshay I resume (1)' and 'Akshay I resume (2)'. The 'File name' field shows 'Akshay I resume (1)'. The 'Open' button is highlighted. In the background, an 'Upload API' interface is visible, showing a successful upload of a file named 'MohanasriD.pdf' with a size of 94 ms and 448 B.

The bottom screenshot shows the 'Upload API' interface with a 'POST Phase 2 - PDF Upload' request. The URL is 'http://localhost:3000/v1/resume/upload'. The request body is a JSON object: 

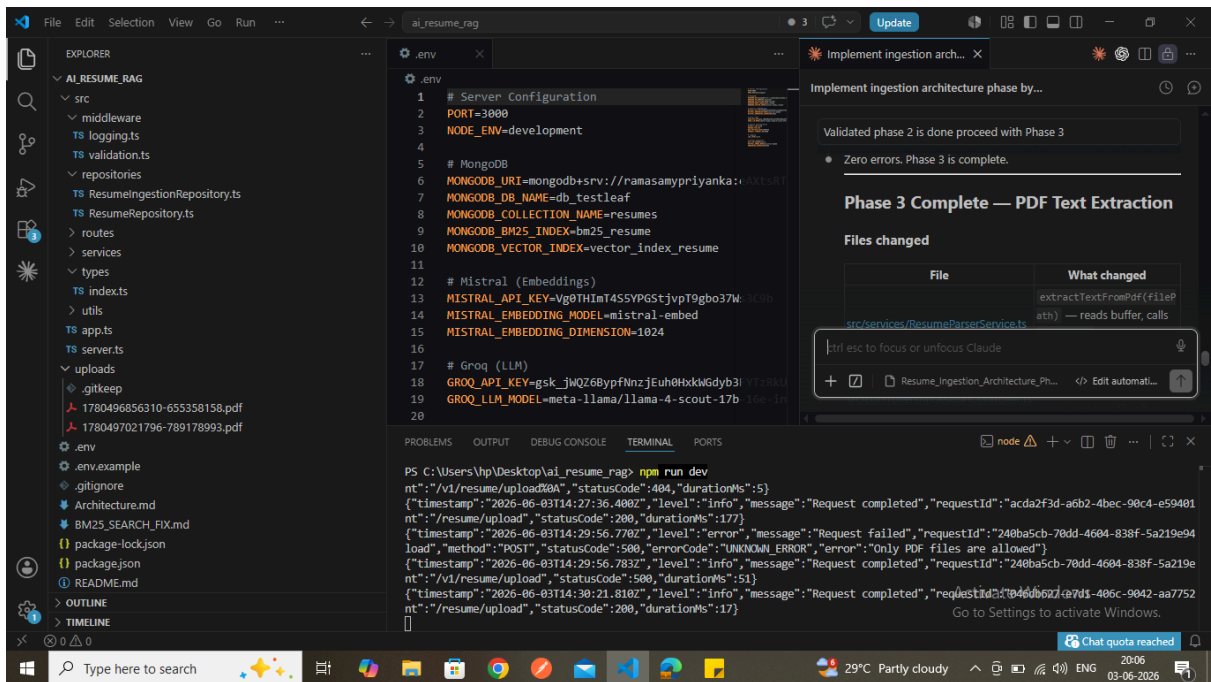
```
{ "requestId": "240ba5cb-78dd-4684-838f-5a219e94fec2", "error": { "code": "UNKNOWN_ERROR", "message": "Only PDF files are allowed" }, "timestamp": "2026-06-03T14:29:56.788Z" }
```

. The response status is '500 Internal Server Error' with a size of 119 ms and 418 B.

Files were available in upload folder

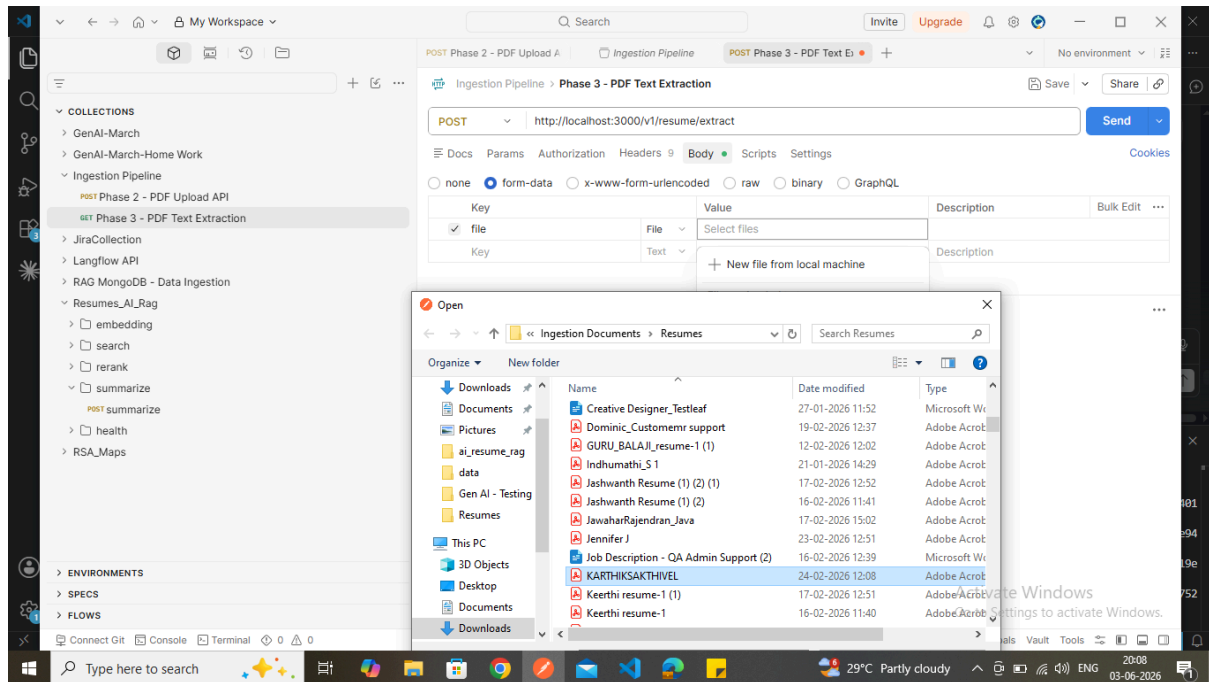


## Phase 3 — PDF Text Extraction.

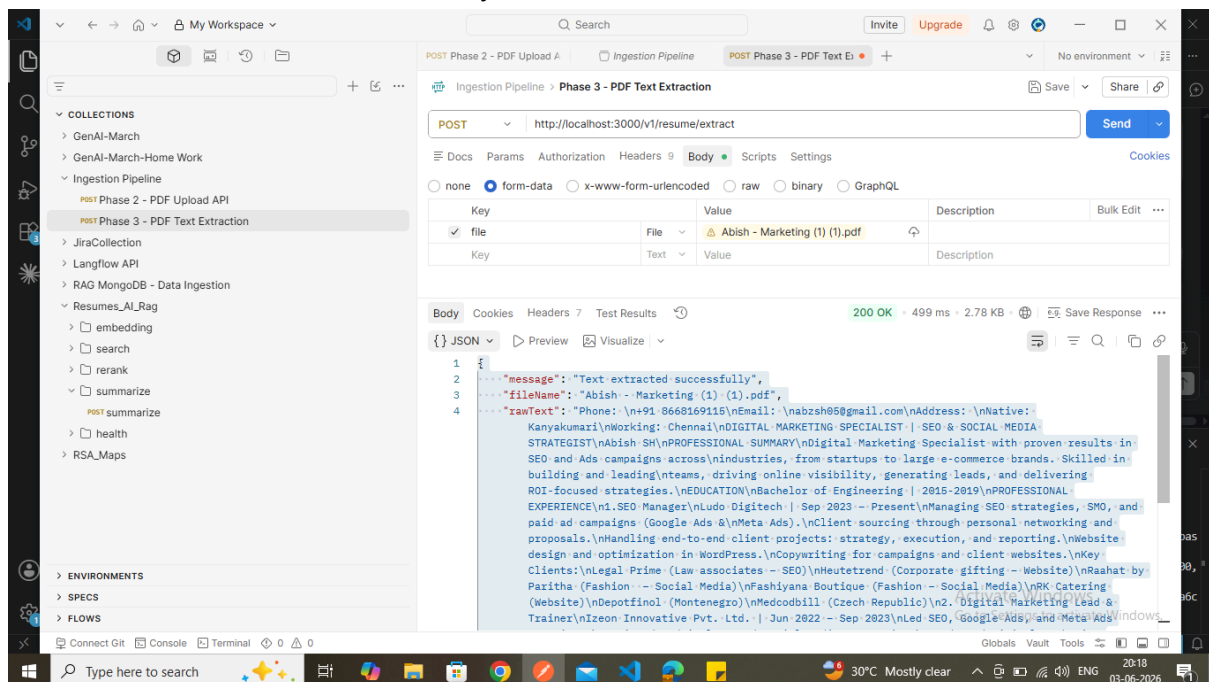




## Validating Phase 3 in PostMan



## Data extracted from PDF successfully



## Phase 4 : Text Cleaning

VS Code interface showing the implementation of Phase 4: Text Cleaning. The Explorer panel displays the project structure for 'ai\_resume\_rag', including files like 'logging.ts', 'validation.ts', 'ResumeRepository.ts', 'routes', 'services', 'types', 'index.ts', 'utils', 'app.ts', 'server.ts', 'uploads', '.gitkeep', 'BM25\_SEARCH\_FIX.md', 'package-lock.json', 'package.json', 'README.md', 'OUTLINE', and 'TIMELINE'. The .env file is configured with the following settings:

```
.env
1 # Server Configuration
2 PORT=3000
3 NODE_ENV=development
4
5 # MongoDB
6 MONGODB_URI=mongodb+srv://ramasamypriyanka:DataR@
7 MONGODB_DB=db_testleaf
8 MONGODB_COLLECTION=resumes
9 MONGODB_BM25_INDEX=bm25_resume
10 MONGODB_VECTOR_INDEX=vector_index_resume
11
12 # Mistral (Embeddings)
13 MISTRAL_API_KEY=Vg0THiMt4S5YPGstjvT9gbo37Mj9C0b
14 MISTRAL_EMBEDDING_MODEL=mistral-embed
15 MISTRAL_EMBEDDING_DIMENSION=1024
16
17 # Groq (LLM)
18 GROQ_API_KEY=gsk_jWQ26BypfNnzjEuh0HxkGdyb3IvTz8NM
19 GROQ_LLM_MODEL=meta-llama-4-scout-17b-16e-1n
20
```

The terminal shows the server starting successfully:

```
PS C:\Users\Vhp\Desktop\ai_resume_rag> npm run dev
[nodemon] starting 'ts-node src/server.ts'
[nodemon] restarting due to changes...
[nodemon] starting 'ts-node src/server.ts'
[nodemon] restarting due to changes...
[nodemon] starting 'ts-node src/server.ts'
{"timestamp":"2026-06-03T14:49:48.927Z","level":"info","message":"MongoDB connected successfully","component":"MongoDB","database":"db_testleaf"}
{"timestamp":"2026-06-03T14:49:49.175Z","level":"info","message":"Server started successfully on port 3000","port":3000,"nodeEnv":"development","mongodb":"db_testleaf"}

```

A sidebar on the right indicates 'Phase 4 Complete — Text Cleaning' with a list of files changed:

| File           | What changed           |
|----------------|------------------------|
| cleanText(raw) | normalizes line breaks |

## Validating the text cleaning via PostMan

Postman interface showing a POST request to 'http://localhost:3000/v1/resume/clean'. The request body is a JSON object:

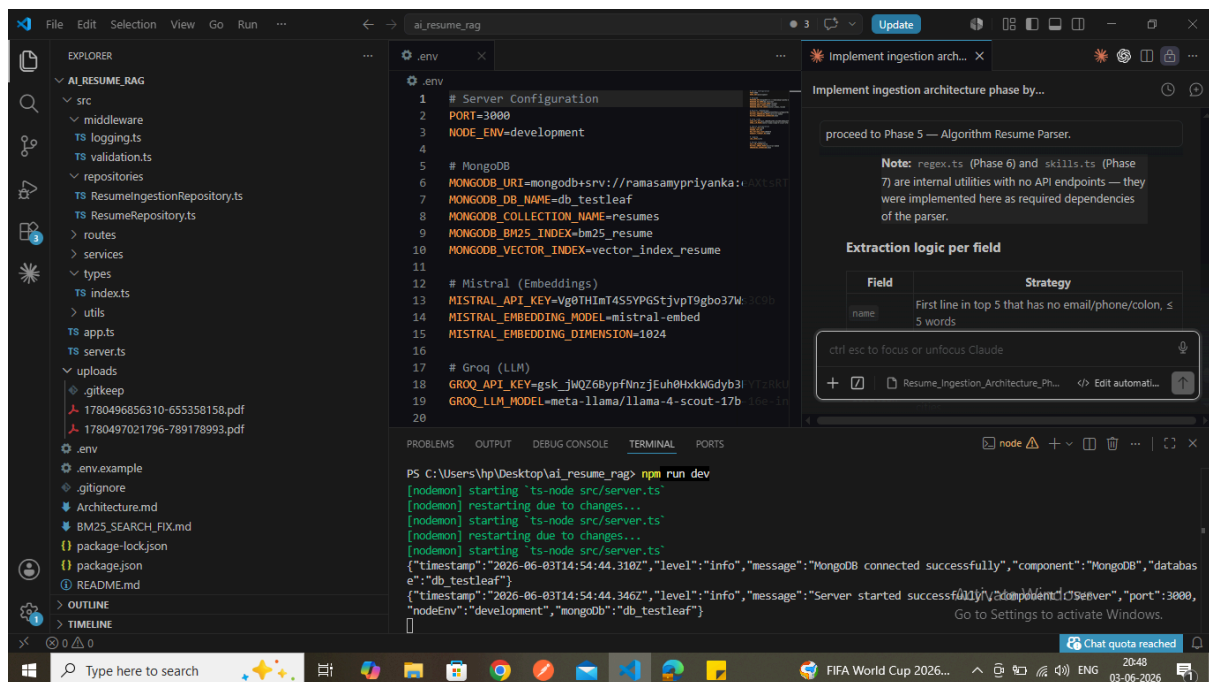
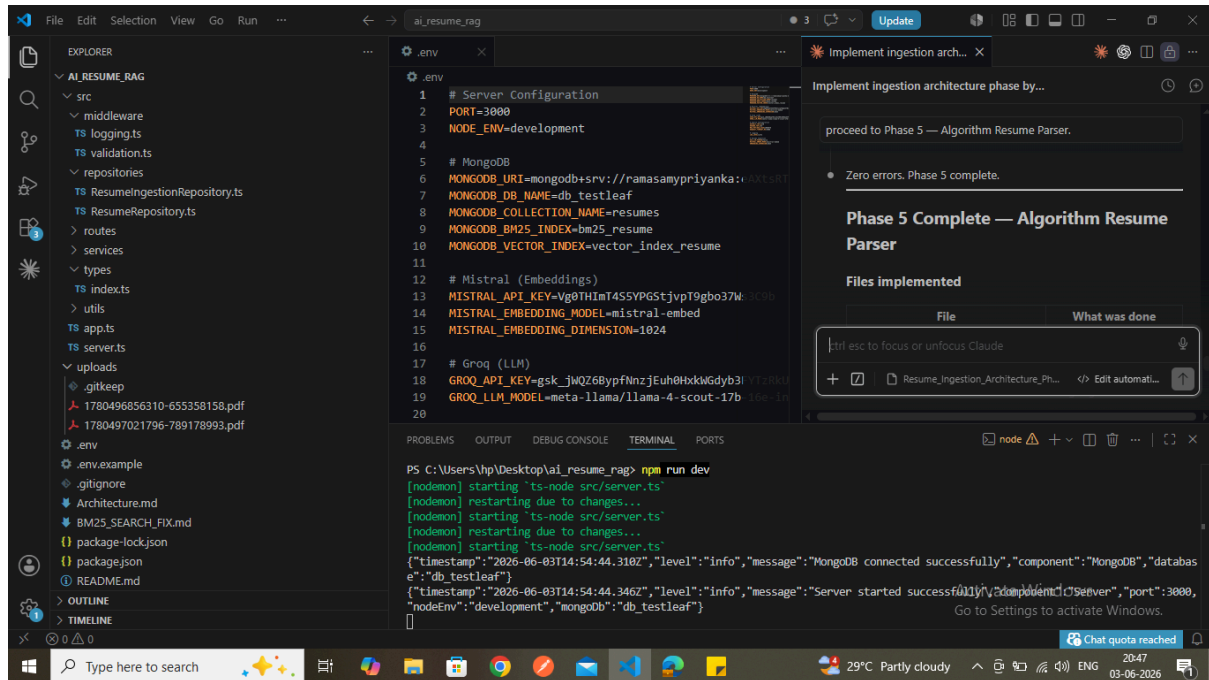
```
{
  "file": "Anish_Kumar_S_Resume_1 (1).pdf"
}
```

The response is a 200 OK status with a JSON body:

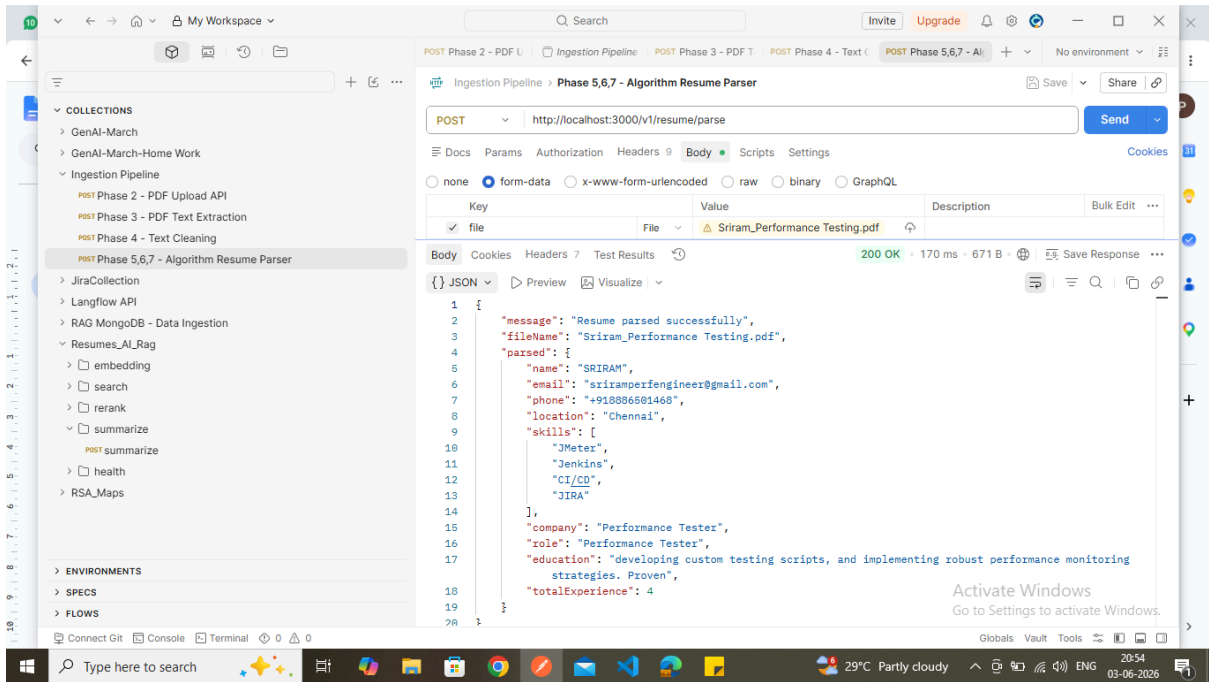
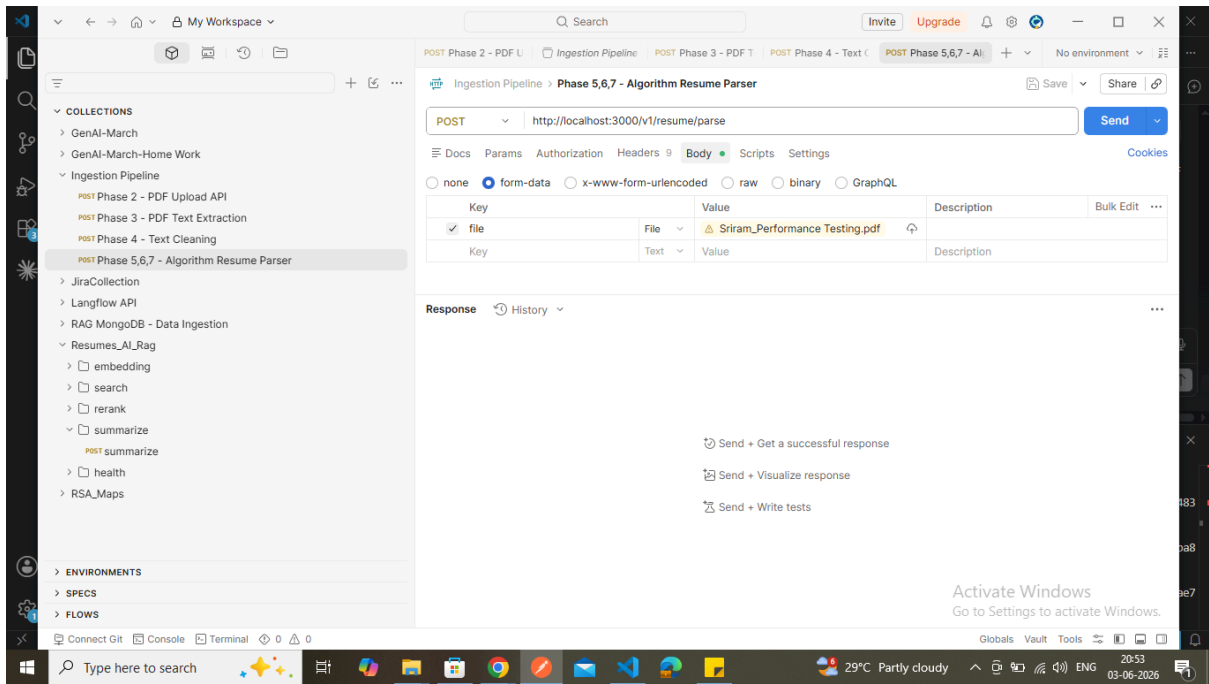
```
{
  "message": "Text cleaned successfully",
  "fileName": "Anish_Kumar_S_Resume_1 (1).pdf",
  "rawText": "ANISH KUMAR S \nQuality Assurance Analyst \nanishkumarsankar2023@gmail.com | +91 93633 84114 \nChennai TN \n\nPROFESSIONAL SUMMARY \n4 years of industry experience in the area of Software Testing (Manual, Web, \nMobile and API) with a solid understanding of Test Planning, Test Design, Test \nExecution and Defect Reporting & Tracking \nExpertise in Understanding and Analyzing Test Requirements, Tracking changes \nand maintenance of Test Requirements \nStrong Experience in Automating Web Application Testing using Selenium \nWebDriver with Cucumber framework \nStrong Experience in Java Programming, Junit and Testing \nWell versed with Handling Elements in Selenium WebDriver \nStrong Experience writing Test cases using Element locators, WebDriver methods \nand Java programming features \nStrong Experience in handling tools like GIT/HUB \nExecuting Selenium Test Cases and Reporting defects \nExperience in developing test cases based on requirements in JIRA \nExperience in POSTMAN Tool and Rest Assured for API Integration \nGood experience using Appium for iOS and Android applications. \nExperience in executing queries using MYSQL \nGood written, communication, interpersonal skills, proven team player with an \nanalytical mind bent to problem solving and delivering"
}
```

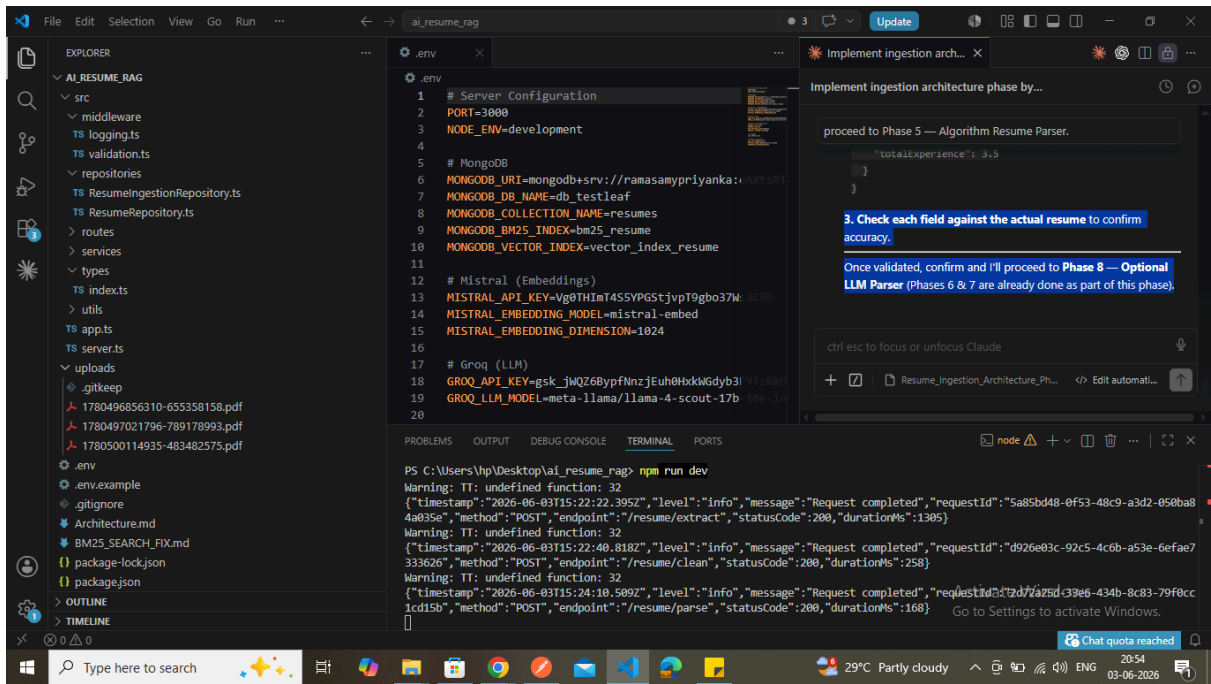


## Phase 5 ,6,7 — Algorithm Resume Parser

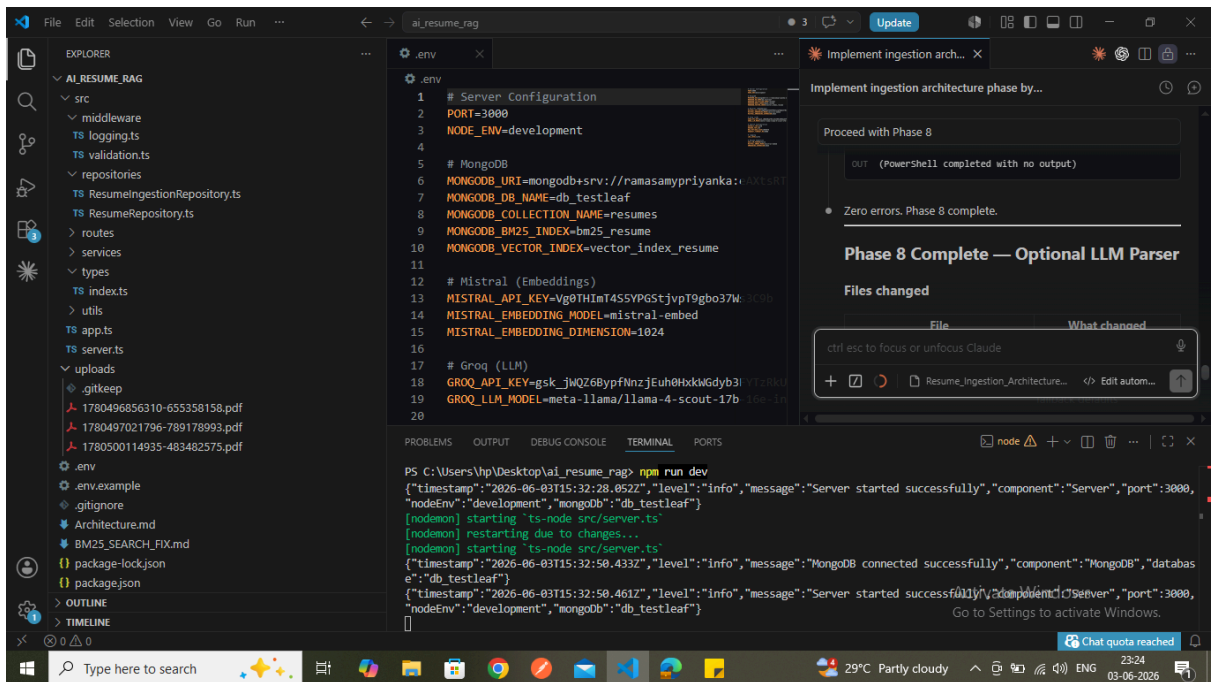


## Validating Phase 5 in Postman





## Phase 8 - Optional LLM Parser



## Validating in Postman

My Workspace

Search

Invited Upgrade

POST Phase 2 - PDF Upload API POST Phase 3 - PDF Text Extraction POST Phase 4 - Text Cleaning POST Phase 5,6,7 - Algorithm Resume Parser POST Phase 8 - LLM Parser

Ingestion Pipeline > Phase 8 - LLM Parser

POST http://localhost:3000/v1/resume/llm-parse

Docs Params Authorization Headers 9 Body Scripts Settings Cookies

☐ none ☒ form-data ☐ x-www-form-urlencoded ☐ raw ☐ binary ☐ GraphQL

| Key  | Value                          | Description | Bulk Edit   |
|------|--------------------------------|-------------|-------------|
| file | Sriram_Performance Testing.pdf |             |             |
| Key  | Text                           | Value       | Description |

Body Cookies Headers 7 Test Results 200 OK 1.38 s 692 B Save Response

JSON Preview Visualize

```
1 {
2   "message": "Resume parsed successfully",
3   "fileName": "Sriram_Performance Testing.pdf",
4   "parser": "algorithm",
5   "parsed": {
6     "name": "SRIRAM",
7     "email": "sriramperfengineer@gmail.com",
8     "phone": "+918886501468",
9     "location": "Chennai",
10    "skills": [
11      "JMeter",
12      "Jenkins",
13      "CI/CD",
14      "JIRA"
15    ],
16    "company": "Performance Tester",
17    "role": "Performance Tester",
18  }
19 }
```

Activate Windows  
Go to Settings to activate Windows.

Globals Vault Tools

File Edit Selection View Go Run

ai\_resume\_rag

Update

Implement ingestion arch...

Implement ingestion architecture phase by...

Proceed with Phase 8

2. Test with USE\_LLM\_PARSER=true

In .env set USE\_LLM\_PARSER=true → nodemon restarts → send the same request.

Expected response:

```
{
  "message": "Resume parsed successfully",
  "fileName": "resume.pdf",
}
```

ctrl esc to focus or unfocus Claude

+ .env Edit automatically

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

node

```
PS C:\Users\Vhp\Desktop\ai_resume_rag> npm run dev
[nodemon] restarting due to changes...
[nodemon] starting "ts-node src/server.ts"
{"timestamp":"2026-06-03T15:32:50.433Z","level":"info","message":"MongoDB connected successfully","component":"MongoDB","database":"db_testleaf"}
{"timestamp":"2026-06-03T15:32:50.461Z","level":"info","message":"Server started successfully","component":"Server","port":3000,"nodeEnv":"development","mongodb":"db_testleaf"}
Warning: ITs: undefined function: 32
{"timestamp":"2026-06-03T17:55:46.338Z","level":"info","message":"Request completed","requestId":"3019088d-41ef-43d6-3d582466d91b","method":"POST","endpoint":"/resume/llm-parse","statusCode":200,"durationMs":964}
Go to Settings to activate Windows.
```

Ln 31, Col 20 Spaces: 4 UTF-8 CRLF Dotenv Chat quota reached

## Phase 9 — Mistral Embedding Generation.

EXPLORER

- AI\_RESUME\_RAG
  - src
    - middleware
      - logging.ts
      - validation.ts
    - repositories
      - ResumeIngestionRepository.ts
      - ResumeRepository.ts
    - routes
    - services
    - types
      - index.ts
    - utils
    - app.ts
    - server.ts
    - uploads
    - .gitkeep
    - 1780496856310-655358158.pdf
    - 1780497021796-789178993.pdf
    - 1780500114935-483482575.pdf

.env

```
18 GROQ_API_KEY=gsk_jW0Z6BypfNnzjEuh0HbkkwGdyb3l
19 GROQ_LLM_MODEL=meta-llama/llama-4-scout-17b
20
21 # Search Configuration
22 DEFAULT_TOP_K=10
23 RERANK_TOP_K=8
24 MAX_PAYLOAD_SIZE=1048576
25 REQUEST_TIMEOUT_MS=5000
26
27 # Logging
28 LOG_LEVEL=info
29
30 # Resume Ingestion
31 USE_LLM_PARSER=true
32 MISTRAL_EMBED_MODEL=mistral-embed
33 EMBEDDING_DIMENSION=1024
34
```

Implement ingestion architecture phase by...

Proceed with Phase 9

Zero errors. Phase 9 complete.

Phase 9 Complete — Mistral Embedding Generation

Files changed

ctrl esc to focus or unfocus Claude

node

PS C:\Users\hvp\Desktop\ai\_resume\_rag> npm run dev
ed145c", "method": "POST", "endpoint": "/resume/llm-parse", "statusCode": 200, "durationMs": 279}
[nodeemon] restarting due to changes...
[nodeemon] starting "ts-node src/server.ts"
[nodeemon] restarting due to changes...
[nodeemon] starting "ts-node src/server.ts"
{"timestamp": "2026-06-03T18:00:05.099Z", "level": "info", "message": "MongoDB connected successfully", "component": "MongoDB", "database": "db\_testleaf"}
{"timestamp": "2026-06-03T18:00:05.121Z", "level": "info", "message": "Server started successfully", "component": "Server", "port": 3000, "nodeEnv": "development", "mongodb": "db\_testleaf"}
}

My Workspace

Search

Invited Upgrade

POST Phase 2 POST Phase 3 POST Phase 4 POST Phase 5 POST Phase 6 POST Phase 7 POST Phase 8 POST Phase 9

Ingestion Pipeline > Phase 9 - Mistral Embedding Generation

POST http://localhost:3000/v1/resume/embed

Docs Params Authorization Headers 0 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL

| Key  | Value                          | Description | Bulk Edit   |
|------|--------------------------------|-------------|-------------|
| file | Sriram_Performance Testing.pdf |             |             |
| Key  | Text                           | Value       | Description |

Body Cookies Headers 7 Test Results 200 OK 2.34 s 19.21 KB Save Response

JSON Preview Visualize

```
1 {
2   "message": "Embedding generated successfully",
3   "fileName": "Sriram_Performance Testing.pdf",
4   "embeddingModel": "mistral-embed",
5   "embeddingDimension": 1024,
6   "embedding": [
7     -0.0617175674438476562,
8     0.0310211181648625,
9     0.036560085859375,
10    -0.0609255409240722656,
11    0.0194091796875,
12    0.0307464599609375,
13    0.0177154541015625,
14    -0.067259368096484375,
15    0.004817962646484375,
16    0.0212554931640625,
17    -0.0622144317626953125,
```

Activate Windows  
Go to Settings to activate Windows.

## Phase 10 — MongoDB Ingestion.

The screenshot shows the VS Code editor with the `ai_resume_rag` project. The `.env` file is open, showing configuration for the MongoDB ingestion phase. The terminal displays the output of `npm run dev`, indicating that the server has started successfully and is connected to MongoDB.

```
.env
GROQ_API_KEY=gsk_jwQ26BypfNnzjEuh0HbdkwGdyb3l
GROQ_LLM_MODEL=meta-llama/llama-4-scout-17b
# Search Configuration
DEFAULT_TOP_K=10
RERANK_TOP_K=8
MAX_PAYLOAD_SIZE=1048576
REQUEST_TIMEOUT_MS=5000
# Logging
LOG_LEVEL=info
# Resume Ingestion
USE_LLM_PARSER=true
MISTRAL_EMBED_MODEL=mistral-embed
EMBEDDING_DIMENSION=1024
```

Terminal Output:

```
PS C:\Users\hvp\Desktop\ai_resume_rag> npm run dev
"nodeEnv":"development","mongoDb":"db_testleaf")
[nodemon] restarting due to changes...
[nodemon] starting "ts-node src/server.ts"
[nodemon] restarting due to changes...
[nodemon] starting "ts-node src/server.ts"
{"timestamp":"2026-06-03T18:04:38.100Z","level":"info","message":"MongoDB connected successfully","component":"MongoDB","database":"db_testleaf"}
{"timestamp":"2026-06-03T18:04:38.166Z","level":"info","message":"Server started successfully, component: Server, port: 3000, nodeEnv: development, mongoDb: db_testleaf"}
```

The screenshot shows the Ingestion Pipeline interface for Phase 10 - MongoDB Ingestion. The pipeline is configured to ingest data from a PDF file into a MongoDB database. The JSON body of the POST request is displayed, showing the resume details and the embedding dimension.

POST Phase: Ingestion Pipeline > Phase 10 - MongoDB Ingestion

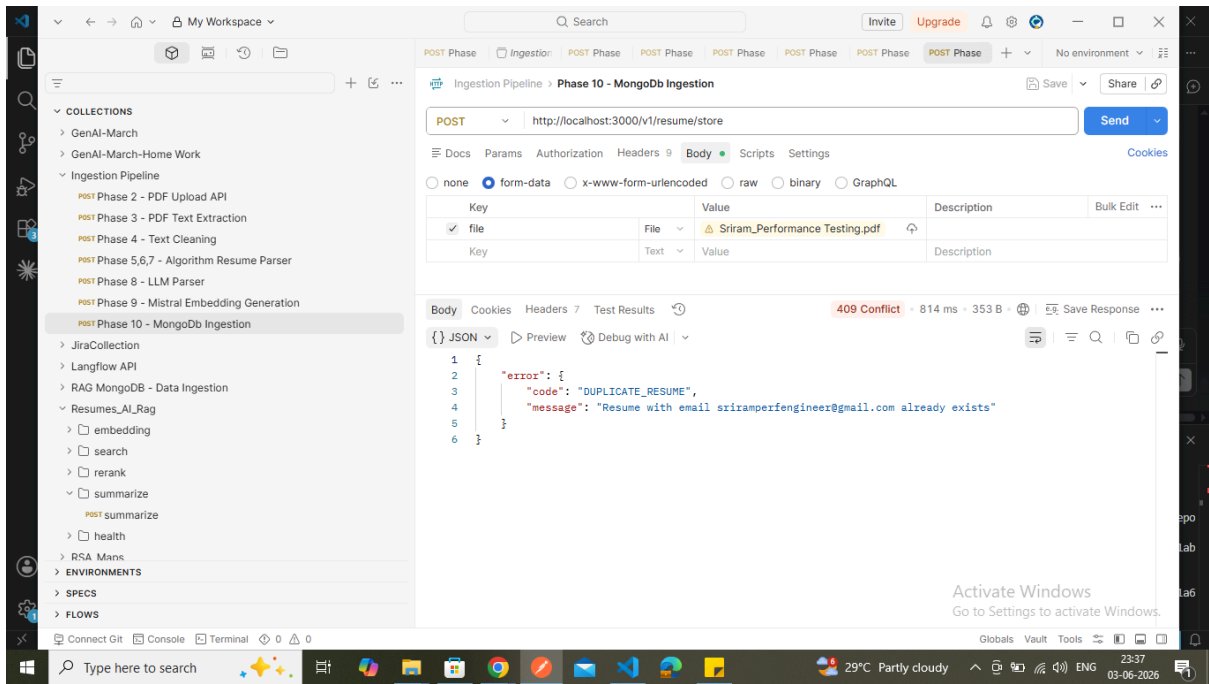
URL: `http://localhost:3000/v1/resume/store`

Body:

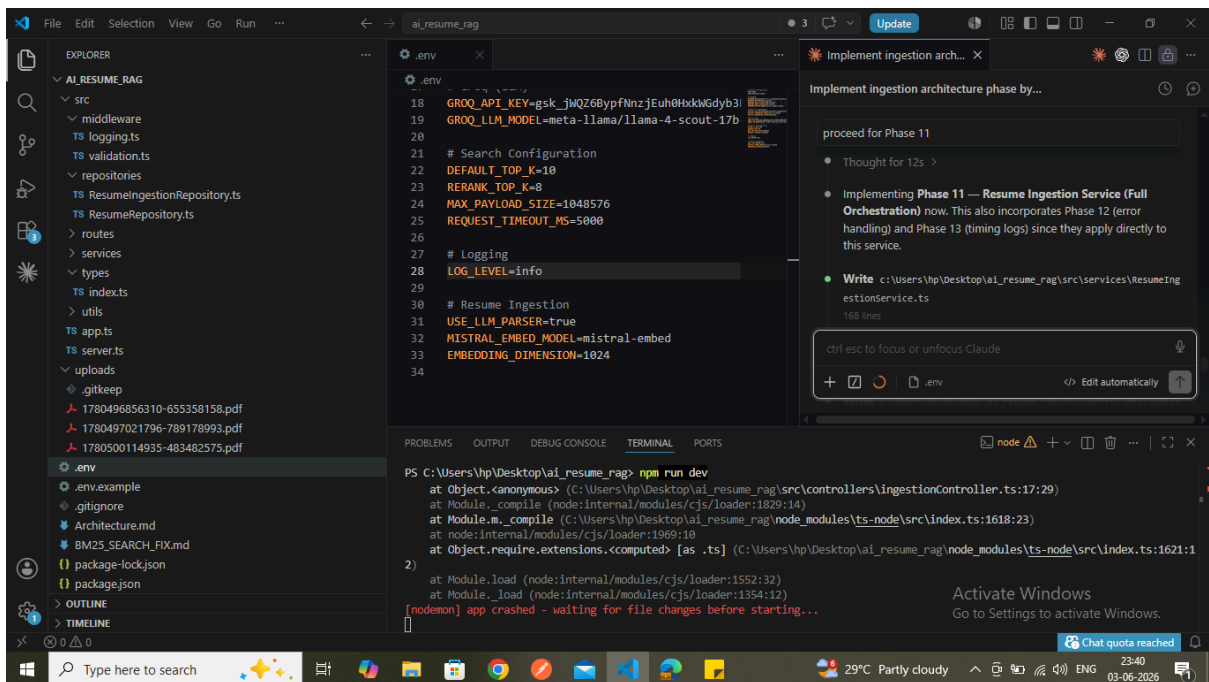
```
{
  "message": "Resume stored successfully",
  "resumeId": "6a206d4b1a4e31b93ecf38ad",
  "fileName": "Sriram_Performance Testing.pdf",
  "name": "SRIRAM",
  "email": "sriramperfeengineer@gmail.com",
  "skills": [
    "JMeter",
    "Blaze Meter",
    "Dynatrace",
    "JIRA",
    "Microsoft Office",
    "Jenkins"
  ],
  "embeddingDimension": 1024
}
```

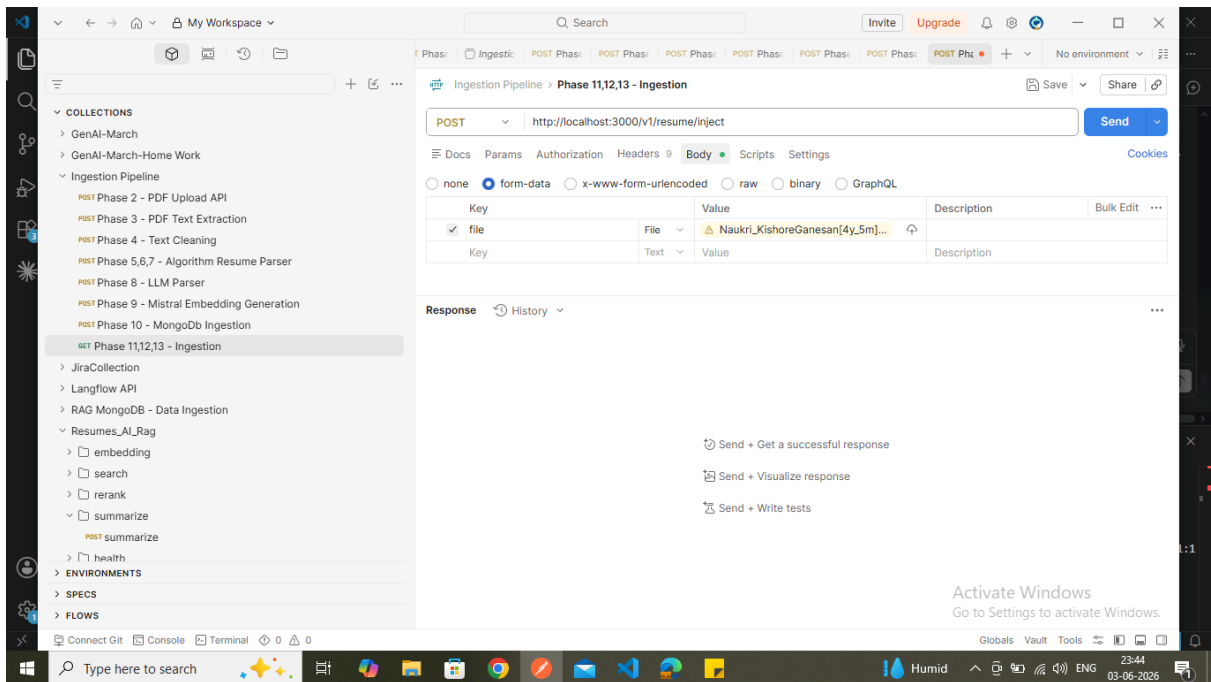
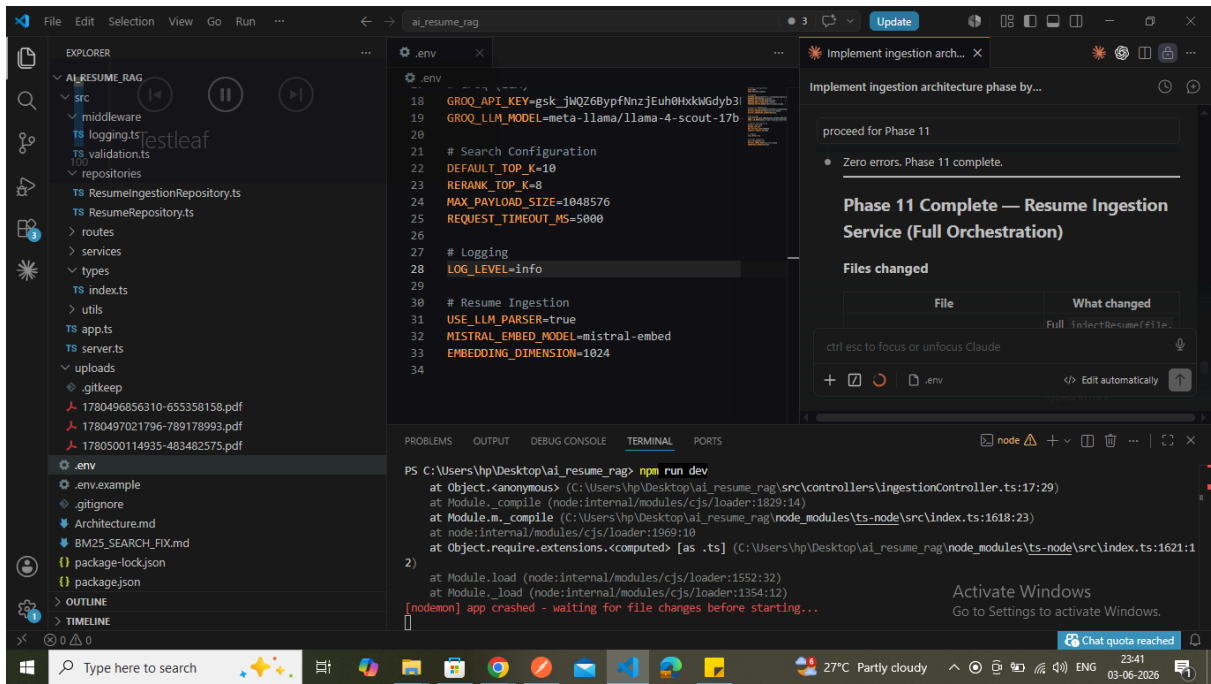
Uploading same resume and checks for error





## Phase 11,12,13 Resume Ingestion Service (Full Orchestration)





My Workspace

Search

Invites Upgrade

Phases: Ingestion POST Phase POST Phase POST Phase POST Phase POST Phase POST Phase POST Phase POST Phase

Ingestion Pipeline > Phase 11,12,13 - Ingestion

POST http://localhost:3000/v1/resume/inject

Docs Params Authorization Headers 9 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL

| Key  | Value                           | Description | Bulk Edit   |
|------|---------------------------------|-------------|-------------|
| file | Naukri_KishoreGanesan[4y_5m]... |             |             |
| Key  | Text                            | Value       | Description |

Body Cookies Headers 7 Test Results 201 Created 4.41 s 722 B Save Response

```
{
  "message": "Resume injected successfully",
  "requestId": "9cd9ce54-63b4-48f8-9a25-8e82288e79db",
  "resumeId": "6a286f7a93594823f3663368",
  "fileName": "Naukri_KishoreGanesan[4y_5m].pdf",
  "name": "KISHORE GANESAN",
  "email": "kishoreganesan1486@gmail.com",
  "skills": [
    "Apache JMeter",
    "Neoload",
    "Fiddler",
    "Postman",
    "Grafana",
    "Influx DB",
    "Jpos",
    "Wireshark",
    "Jenkins"
  ]
}
```

Activate Windows  
Go to Settings to activate Windows.

Globals Vault Tools

23:46  
03-06-2026

My Workspace

Search

Invites Upgrade

Phases: Ingestion POST Phase POST Phase POST Phase POST Phase POST Phase POST Phase POST Phase

Ingestion Pipeline > Phase 11,12,13 - Ingestion

POST http://localhost:3000/v1/resume/inject

Docs Params Authorization Headers 9 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL

| Key  | Value                           | Description | Bulk Edit   |
|------|---------------------------------|-------------|-------------|
| file | Naukri_KishoreGanesan[4y_5m]... |             |             |
| Key  | Text                            | Value       | Description |

Body Cookies Headers 7 Test Results 201 Created 4.41 s 722 B Save Response

```
{
  "Postman",
  "Grafana",
  "Influx DB",
  "Jpos",
  "Wireshark",
  "Jenkins",
  "Docker"
},
  "embeddingDimension": 1024,
  "timings": {
    "extractMs": 740,
    "parseMs": 2446,
    "embeddingMs": 1023,
    "mongoInsertMs": 62,
    "totalMs": 4318
  }
}
```

Activate Windows  
Go to Settings to activate Windows.

Globals Vault Tools

27°C Partly cloudy  
23:46  
03-06-2026

## Duplicate resume uploaded and thrown error

The screenshot displays a web application interface for managing ingestion pipelines. On the left, a sidebar lists various collections and environments. The main panel shows the configuration for 'Phase 11,12,13 - Ingestion', which is a POST request to 'http://localhost:3000/v1/resume/inject'. The request body is set to 'form-data'. A table below the configuration shows a file named 'Naukri\_Vishalini[0y\_0m].pdf' being uploaded. The response section shows a '409 Conflict' error with the following JSON body:

```
{  "requestId": "3b43b8d5-b1b3-47f8-b1e8-a7a1b1bac8d2",  "error": {    "code": "DUPLICATE_RESUME",    "message": "Resume with email vishalini1462@gmail.com already exists"  },  "timestamp": "2026-06-04T13:36:38.515Z"}
```

## Uploaded .docx file and thrown an error

The screenshot displays the same web application interface, but now showing a '500 Internal Server Error' for uploading a .docx file. The request body is still 'form-data', but the file is named 'Naukri\_SathishKumar[8y\_0m].docx'. The response section shows a '500 Internal Server Error' with the following JSON body:

```
{  "requestId": "902f9965-3c03-4539-90f1-07e5792bc624",  "error": {    "code": "UNKNOWN_ERROR",    "message": "Only PDF files are allowed"  },  "timestamp": "2026-06-04T13:38:58.385Z"}
```

## End to End Search

The screenshot displays a web application interface for managing an ingestion pipeline. The sidebar on the left lists various collections under 'My Workspace', including 'GenAI-March', 'GenAI-March-Home Work', and 'Ingestion Pipeline'. The main panel shows the configuration for the 'end to end search' pipeline, which is a POST request to 'http://localhost:3000/v1/search'. The request body is a JSON object with the following structure:

```
1 {
2   "query": "Java QA Engineer Selenium",
3   "topK": 5,
4   "summarize": false
5 }
```

The response viewer shows a successful 200 OK status with a response time of 2.38 s and a size of 12.74 KB. The response is a JSON object with the following structure:

```
1 {
2   "requestId": "7a6b48f7-1b84-4c88-a232-569fd7701587",
3   "results": [
4     {
5       "_id": "6a1d7c18afb3e44c811af1d5",
6       "name": "Lakshmi G",
7       "role": "Test Manager / Scrum Master",
8       "skills": ["Selenium", "Cucumber", "TestNG", "Agile", "Scrum", "Test Management", "QA Strategy", "Functional Testing", "Backend Testing", "API Testing", "Automation Testing"],
9       "resumeId": "6a1d7c18afb3e44c811af1d5",
10      "score": 2.546985994415283,
11      "snippet": "Lakshmi G | lakshmi1@gmail.com | +91 9563456898 | Chennai, Tamil Nadu"
12    }
13   ]
14 }
```

The response also includes a 'PROFESSIONAL SUMMARY' section:

PROFESSIONAL SUMMARY Dynamic Test Manager and Certified Scrum Master with 12+ years of QA, testing, and project management experience in the IT industry. Proven expertise in leading testing teams, managing client communications, driving Agile Scrum practices, and ensuring high-quality project delivery. Hands-on experience in functional, backend, and